

Mathematical Closure Program

Status: **Draft program architecture**

Purpose

Turn rigorous scale-limit analysis into a computational program for detecting, representing, and correcting closure failure. The program connects two kinds of evidence:

1. theorem-faithful canonical systems in which microscopic-to-kinetic and kinetic-to-fluid limits can be tested numerically; and
2. TEMPEST application profiles in which the assumptions behind those limits fail and an augmented closure is needed.

The application layer contains two complementary profiles. Dusty plasma makes eliminated fields, strong correlations, context, and memory observable in particle-resolved data. Core-collapse supernovae test a different reduction: whether unresolved 3D shock-turbulence dynamics can be represented self-consistently in a reduced 1D carrier.

Mathematical foundations

Three papers motivate one connected program:

- Deng, Hani, and Ma, *Long time derivation of the Boltzmann equation from hard sphere dynamics*, organize departure from particle independence through cumulants, time layers, partial time expansions, and collision-history diagrams called molecules.
- Deng, Hani, and Ma, *Hilbert's sixth problem: derivation of fluid equations via Boltzmann's kinetic theory*, connect the particle-to-Boltzmann limit to compressible Euler and incompressible Navier-Stokes-Fourier limits under stated scalings and preparation assumptions.
- Deng and Hani, *Long time justification of wave turbulence theory*, use time-layered spectral cumulants and canonical layered gardens to justify the wave kinetic equation over its full lifespan.

These are not ready-made plasma or turbulence closures. They provide a mathematical organization of what a reduced description discards and a way to state when that discarded information remains negligible.

Hierarchy and closure defect

Let $(F_N(Z_N, t))$ denote an ensemble density for a microscopic (N)-body system and $(f_s(z_1, \dots, z_s, t))$ its (s)-particle marginals. At second order, write

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$$f_2(z_1, z_2, t)$$

$$f_1(z_1, t)f_1(z_2, t) + g_2(z_1, z_2, t),]$$

where (g_2) is the connected two-particle correlation. Higher connected cumulants $(g_3, g_4, \dots)$ measure departures from lower-order factorization.

A generic first-level reduced equation can be organized as

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partial f_1

$$\mathcal{L}[f_1] + \mathcal{C}[f_1] + \mathcal{R}[g_2, g_3, \dots].$$

The classical closure either neglects $\mathcal{R}$ or absorbs its asymptotic effect into $\mathcal{C}$. The computational target is not the full high-dimensional cumulant hierarchy. It is the smallest representation of $\mathcal{R}$ needed to predict selected resolved observables over the desired time horizon.

For basis or test functions (ψ_i) , projected statistics include

$$[a_i = \mathbb{E}[\psi_i(Z)], \quad \text{quad}$$

κ_{ij}

$$\mathbb{E}[\psi_i(Z)\psi_j(Z)] - a_i a_j,$$

with analogous higher connected cumulants. Candidate compression mechanisms include spectral bases, low-rank tensors, random sketches, collision-graph features, and learned encoders.

Time-layered computational state

The long-time proofs avoid expanding the leading solution from the current time all the way back to $(t=0)$. They restart short-time expansions at later base times while propagating structured information about connected correlations.

A computational analogue is

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S_{ell}

$$\text{left}(f_1^{\ell}, P_{\ell} g_2^{\ell}, P_{\ell} g_3^{\ell}, \text{hellright}), \quad \text{quad } S_{\ell+1} = \text{Phitheta}(S_{\ell}; \mu), \quad]$$

where (P_{ℓ}) is a controlled projection, (hell) is a finite history state, and (μ) contains scaling and physical parameters. The baseline kinetic or fluid evolution remains explicit; $(\text{Phi}_{\text{theta}})$ supplies only the unresolved cumulant or memory contribution.

Central hypotheses

The program tests four falsifiable hypotheses:

1. **Detectability:** projected cumulants or interaction-history features identify the onset of closure failure before large resolved-state error.
2. **Compressibility:** the dynamically important closure defect occupies a low-dimensional or otherwise tractable representation.
3. **Finite memory:** a bounded history state materially improves long-time prediction over a Markov closure without growing indefinitely.
4. **Transfer:** the representation and diagnostics learned in canonical and dusty-plasma systems remain useful, after physics-specific adaptation, for a core-collapse-supernova turbulence closure.

Failure of any hypothesis is a reportable scientific outcome and triggers a stop, pivot, or restriction of scope.

Benchmark architecture

Track A: canonical mathematical validation

A1. Particle-to-Boltzmann

- Microscopic reference: event-driven hard-disk or hard-sphere ensembles.
- Reduced reference: a Boltzmann solver under matching scaling and initial data.
- Measured losses: factorization error, projected cumulants, collision-history connectivity, recollision counts, and reduced-equation residual.

- Primary question: when and how does propagation of chaos fail at finite parameters and finite time?

A2. NLS-to-wave-kinetic equation

- Microscopic reference: ensemble pseudospectral cubic NLS.
- Reduced reference: a wave kinetic equation solver.
- Measured losses: spectral covariance error, fourth- and sixth-order connected cumulants, resonant-interaction history, and kinetic-equation residual.
- Primary question: what history state is needed beyond a random-phase or Gaussian closure?

A3. Boltzmann-to-fluid

- Kinetic reference: Boltzmann solutions approaching the appropriate hydrodynamic scaling.
- Reduced references: compressible Euler and incompressible Navier-Stokes-Fourier.
- Measured losses: departure from local Maxwellians, Hilbert-expansion remainder, constitutive flux error, and macroscopic limit error.
- Primary question: can a regime indicator detect the boundary of validity of the fluid closure?

Track B: TEMPEST application profiles

B1. Dusty-plasma many-body closure ladder

- Hierarchy: particle tracking or DRIAD to kinetic statistics to moment or coarse collective dynamics.
- Purpose: test transfer to strong correlations, anomalous transport, and experimentally observable particle cumulants.
- Proposed internal ladder:
 1. infer effective pair, environment, graph, or memory-dependent forces after eliminating plasma and wake degrees of freedom;
 2. test the particle-to-kinetic factorization and correlation closure;
 3. test kinetic-to-moment stress, flux, production, and memory closures; and
 4. compare which diagnostics persist across the two reductions.
- Decisive coordinates: coupling and density, screening, charge/mass heterogeneity, forcing history, and the ratio of wake or field relaxation time to resolved particle time.
- Status: retained TEMPEST application profile; production runs remain gated on a scientific owner, reference model, independent validation quantities, and approved parameter sweep.

B2. Core-collapse-supernova turbulence closure

- High-fidelity reference: 3D radiation-hydrodynamic or (M)HD DNS/implicit-LES/LES simulations that resolve or quantify shock-turbulence interaction over a declared filter and resolution hierarchy.
- Reduced carrier: a 1D spherically symmetric supernova model with the minimum thermodynamic, composition, gravity, and neutrino-transport state required by the selected science case.
- Projection: angle or volume averaging of the 3D state onto radial profiles, with Favre/Reynolds conventions and shock tracking declared before data generation.
- Initial target: $[C\{\mathrm{SN}\}^{\mathrm{star}}[X] = \langle F\{3\mathrm{D}\}(X) \rangle_{\Omega} - F\{1\mathrm{D}\}(\langle X \rangle_{\Omega})$,] decomposed into Reynolds stress/turbulent pressure, turbulent energy or enthalpy flux, production, dissipation, mixing, and optional finite-memory terms.
- Candidate augmented state: turbulent kinetic energy, anisotropy or Reynolds-stress coordinates, shock-relative features, filtered cumulants, and a bounded history state.
- Required constraints: total-energy and lepton-number accounting as provided by the carrier, positivity and realizability, stable shock-capturing rollout, causal memory, and recovery of the declared zero-turbulence or calibrated local baseline.
- Application observables: mean shock-radius history, shock revival or runaway classification and timing, turbulent support behind the shock, explosion energy, and nucleosynthesis-sensitive thermodynamic histories, with the version-0 subset fixed by the supernova science owners.
- Status: selected second TEMPEST science application; exact reference simulations, carrier, projections, and acceptance thresholds remain open.

Within B1, the proposed decisive experiment nests an analytic instantaneous pair law, a learned pair kernel, an instantaneous many-particle graph model, and a finite-memory graph or auxiliary-state model. Sweeping wake relaxation, density, and forcing history reveals where pairwise closure fails, where many-particle context is required, and where memory earns its inference cost. Projected cumulants and interaction-history complexity are tested as early warnings of these boundaries. The same ensembles then define kinetic and moment closure defects, converting interaction learning into an input to the next two scale transitions rather than an isolated trajectory-prediction task.

The complete cross-track execution sequence is maintained in `experimental_pipeline.md`.

Closure models to compare

1. Analytic and statistically identifiable effective interaction laws.
2. Contextual or finite-memory graph particle models.
3. Classical Markov kinetic or fluid closure.
4. Explicit finite-cumulant or finite-moment closure.
5. Finite-memory closure using kernels or auxiliary variables.
6. Structure-preserving learned cumulant/memory correction.

7. Direct black-box surrogate as a diagnostic control, not the default model.

PINNs, neural operators, PINOs, structured neural dynamics, and world models remain in the broader bake-off, but they approximate different mathematical objects. The closure and structured-dynamics tracks are the center of this mathematical program. No single scalar ranking should combine force inference, solution approximation, operator learning, closure prediction, and controlled rollout; each is scored against its own declared object and information budget.

Required numerical evidence

- Convergence across the theorem's or model's scaling parameters.
- Ensemble uncertainty and sample-complexity analysis for cumulant estimates.
- A priori closure-defect accuracy and a posteriori embedded-solver stability.
- Long-time factorization, memory, and resolved-observable errors.
- Conservation, positivity, realizability, entropy, and hyperbolicity tests.
- Ablations over cumulant order, projection rank, memory length, and graph features.
- Explicit failure boundaries and comparison with the classical asymptotic limit.

Scope discipline

Do not begin by implementing the proof's full diagrammatic cutting algorithms. Those algorithms establish mathematical bounds and are not automatically efficient numerical closures. First instrument canonical simulators so that cumulants and interaction histories are measurable. Only then test which proof structures are computationally predictive and compressible.